

DAY 4 HOMEWORK

Plan Your First AI Application Before Writing Python

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Project/App Name: Scam Shield

Assignment: Complete all eight planning items for your own AI capstone idea. Do not begin coding until your plan is specific, testable, and easy to explain.

Submission: Submit this completed document and upload it to your GitHub project folder using a clear filename such as day4_ai_app_plan.docx.

1. Clear Problem Statement

Write one specific sentence using this structure: "Many [users] struggle with [specific problem] because [reason]."

Helpful reminder: Describe the problem, not the features of your app. Avoid broad words such as "people," "things," or "everything."

My problem statement:

People receive suspicious emails, text messages, and social media messages and may not know whether they are scams. This app not only checks messages and email for scams but also helps the user learn to recognize scams.

2. User Story

Complete: "As a [type of user], I want to [action], so that [benefit]."

My user story:

As someone who receives many scam emails and messages, I want to create an app that can recognize whether something is a scam or not, so that people don't fall for scams.

3. Input–Process–Output (IPO) Chart

Input is what the user gives the app. **Process** is what the app does. **Output** is what the user receives.

INPUT	PROCESS	OUTPUT
Message: Phone number and text message	AI reads the message and checks for scam warnings, risk level, and generates a simple explanation.	Risk level, Warning signs found, Recommended action.
Email: Sender, Subject, and Body	AI reads the message and checks for scam warnings, risk level, and generates a simple explanation.	Risk level, Warning signs found, Recommended action.

4. Pseudocode

Write your program plan in plain English. Include: START, user input, validation, processing, output, and END. Use IF/ELSE when the app must make a decision.

Step #

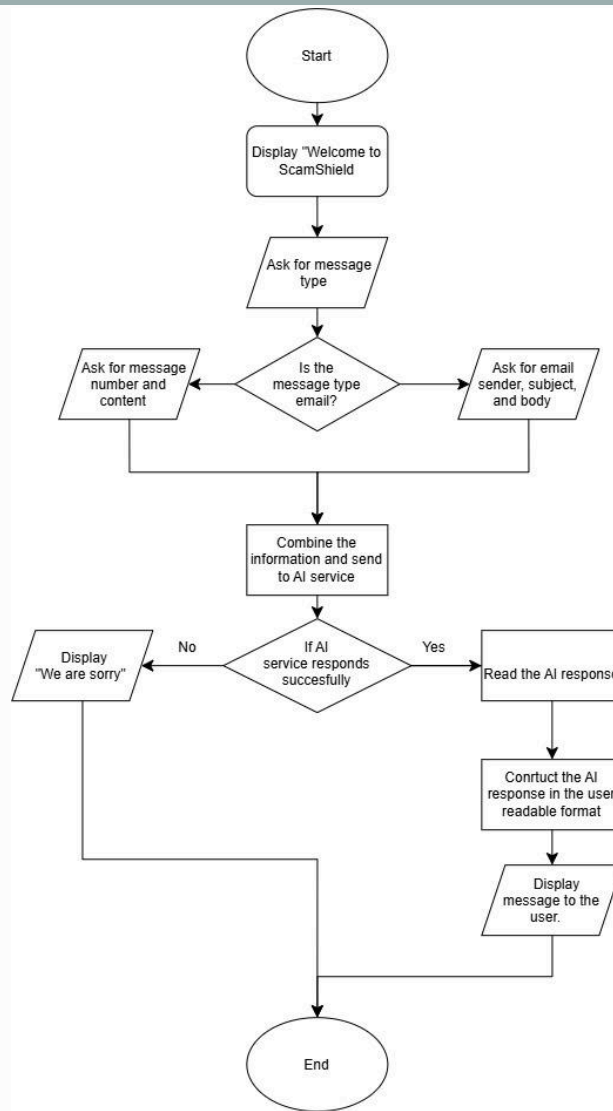
Flowchart Step

- 1 Start
- 2 Display **"Welcome to ScamShield"**
- 3 Ask for message type
- 4 Check: **Is the message type email?**
- 5A If **Yes** → Ask for email sender, subject, and body
- 5B If **No** → Ask for message number and content
- 6 Combine the information
- 7 Send the information to the AI service
- 8 Check: **Did the AI service respond successfully?**
- 9A If **No** → Display **"We are sorry"**
- 9B If **Yes** → Read the AI response
- 10 Convert the AI response into a user-readable format
- 11 Display the message to the user
- 12 End

5. Flowchart

Draw the order of your app. Include at least: Start, Input, one Decision, Process, Output, and End. Label each arrow clearly.

DRAW YOUR FLOWCHART HERE



6. User Interface Sketch

Sketch what the user will see. Include: app title, input areas, button(s), warning/error location, and output area.

SKETCH YOUR APP SCREEN HERE

ScamShield

AI-powered scam message analyzer

Welcome to ScamShield

Enter the details of a suspicious message and we will analyze it for you.

1. Select Message Type

Email

Text Message

2. Enter Details

Email

Text Message

Sender Email

example@example.com

Subject

Enter subject

Body

Enter email body here...

Analyze Message

3. Analysis Result

Your analysis result will appear here after you click 'Analyze Message'.

Tip: Never share personal information or click on suspicious links.

7. Two Test Cases

For each test, show exactly what the user enters and what the app should return. Use one normal test and one edge case, such as empty input, unclear wording, or missing information.

Test Case 1

Test purpose	To check if they can recognize scam emails.
Exact input	Dear Esteemed Friend, My name is Prince Alexander Kamboro, Crown Prince of the Royal Kingdom of Belvaria. I am contacting you personally because I have been informed that you are a trustworthy and honest person who can help my royal family with a VERY URGENT financial situation. Due to political problems in my country, I currently have \$18,500,000 USD that must be transferred outside the country immediately. If you help me receive these funds, I will give you 30% (\$5,550,000) as a reward. THIS OFFER IS 100% GUARANTEED AND COMPLETELY CONFIDENTIAL!!!

	You must respond within 24 HOURS or the government may seize the money and this opportunity will be lost forever. To begin the transfer, please send me the following information immediately: • Your full legal name • Home address • Phone number • Date of birth • Bank name • Bank account number • Routing number • A photo of your driver's license or passport • Your online banking username for verification You will also need to pay a small \$750 international processing fee before the \$18.5 million can be released. The fee must be paid today using gift cards, cryptocurrency, or wire transfer. Please DO NOT contact your bank, police, friends, or family about this transaction. It is a confidential matter involving the Royal Family. Click this secure link immediately to confirm your information: http://royal-belvaria-prize-money.example.invalid/CLAIM-NOW If the link does not work, reply with your bank information directly. IMPORTANT: If you fail to respond today, your name may be removed from the approved beneficiary list and you could face legal investigation for refusing an official royal request. Congratulations in advance. Once I receive your information and payment, the millions of dollars will arrive in your bank account within 2 hours. May God bless you abundantly. His Royal Highness, Prince Alexander Kamboro Royal Kingdom of Belvaria Director of International Royal Finance Email: prince.royal.funds@example.invalid
Expected output	Highest level risk, many warning signs such as an unknown sender, impersonations of royalty, an enormous financial reward, urgency, secrecy, threats,

	grammatical oddities, requests for sensitive information, an upfront fee, gift cards/crypto, suspicious links, and instructions not to contact trusted people or authorities.
How I will know it passed	It declares the email a scam and suggests the user to report it.

Test Case 2

Test purpose	To check if they can tell the difference from a stranger asking for money and someone close to you.
Exact input	Hi Alex, I'm filling out the registration form for your summer program and realized I don't remember the exact date they need for your birthday. Can you send me your full date of birth (month, day, and year) when you get a chance? They also asked me to confirm your full name as it appears on your school records, so please send that too. No rush—I just want to finish the form before I forget. Thanks! Love, Mom
Expected output	Low risk level, shows some warning scans like asking for personal information, suggests checking the sender.
How I will know it passed	Doesn't assume it's automatically a scam.

8. Scope Statement: Version 1

Keep your first version small. List only the most important features under "Will Do." Put future ideas and complicated integrations under "Will Not Do Yet."

VERSION 1 WILL DO	VERSION 1 WILL NOT DO YET
Check emails for scams	Check links and files

One-Breath Project Explanation

Explain your project in 2–3 sentences: who it helps, what information it uses, what it does, and what result it provides.

ScamShield helps everyone who doesn't know how to recognize scam emails and messages and helps them learn how to. It checks if emails and messages are scams and tells the risk level, warning signs, and next steps.

Submission Checklist

- ☐ I completed all eight sections.
- ☐ My problem statement is specific.
- ☐ My input, process, and output match each other.
- ☐ My pseudocode and flowchart show the same sequence.
- ☐ My interface sketch includes input, button, messages, and output.
- ☐ My two test cases include exact input and expected output.
- ☐ My Version 1 scope is small enough to build.
- ☐ I reviewed spelling and clarity.
- ☐ I saved the file with a clear lowercase filename.
- ☐ I uploaded the completed document to GitHub.

Teacher Review (40 points)

Required Item	Points	Teacher Notes
1. Problem statement	/ 5	
2. User story	/ 5	
3. IPO chart	/ 5	
4. Pseudocode	/ 5	
5. Flowchart	/ 5	
6. Interface sketch	/ 5	
7. Two test cases	/ 5	
8. Scope statement	/ 5	